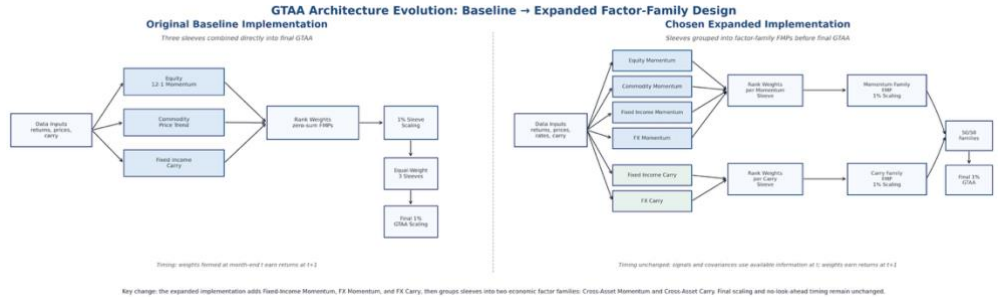


Long/Short Factor-Based GTAA Portfolio

1. Introduction

This report constructs a long/short GTAA portfolio using two economic factor families, Cross-Asset Momentum and Cross-Asset Carry. The baseline strategy uses equity momentum and fixed-income carry. The expanded specification keeps this structure but adds commodity momentum, fixed-income momentum, FX momentum, and FX carry to create a broader cross-asset implementation. The purpose of the expansion is not to chase stronger in-sample performance, but to test whether a wider macro opportunity set can preserve the same risk-controlled construction while improving diversification across equities, commodities, fixed income, and currencies



Each sleeve is first formed as a long/short factor-mimicking portfolio within its own asset class. Sleeve weights are then grouped into the Momentum and Carry families, scaled to 1% annualized volatility, combined with equal family weights, and finally rescaled to a 1% annualized portfolio volatility target. The baseline remains in the report as a benchmark for comparing whether the expanded architecture adds robustness or simply increases complexity, and extended results are available in the accompanying [supplementary report on Github](#).

2. Data Sources

The investment universe spans four asset classes: equities, commodities, fixed income, and foreign exchange. Signals are ranked only within their own asset-class sleeve, rather than pooled across all assets. This avoids comparing signals with different economic units, such as FX interest-rate differentials, bond yield proxies, and commodity price momentum. Each sleeve therefore produces a self-contained long/short FMP, which is later grouped into either the Momentum or Carry family.

Sleeve	Assets	Signal	Source
Equity Momentum	10 country ETFs	12-1 return momentum	Provided project data + Yahoo ETF prices
Commodity Momentum	12 commodity ETFs	12-1 price momentum	Yahoo Finance
Fixed-Income Momentum	10 bond ETFs	12-1 return momentum	Yahoo Finance
FX Momentum	6 currency ETFs	12-1 price momentum	Yahoo Finance
Fixed-Income Carry	10 bond ETFs	yield proxy carry	FRED
FX Carry	6 currency ETFs	foreign 3M rate minus USD 3M rate	FRED

Sub Portfolio Tickers:

Sub-Portfolios	Tickers / series
Equity Momentum	EWA, EWC, EWG, EWJ, EWL, EWQ, EWT, EWU, EWY, SPY
Commodity Momentum	GLD, SLV, USO, UNG, DBA, DBB, PDBC, PALL, PPLT, CORN, WEAT, SOYB
Fixed-Income Momentum /Carry	SHV, SHY, IEI, IEF, TLH, TLT, TIP, BND, LQD, HYG
FX Momentum/Carry	FXE, FXY, FXB, FXA, FXC, FXF

Signals are then ranked within each asset class, and avoids pooling across asset classes, as we can have fundamentally different signal units, wherein, for example, an FX interest rate differential, is not directly comparable to a commodity price momentum. Note that the four asset classes have different data histories, and each part become active at a different point in time, the constraint being, for example, Commodity ETFs only listed in Nov-2004, whereas FX ETFs listed from 2005.

Factors, Methodology, and Portfolio Construction

2.1 Signals

Equity and Fixed-Income Momentum

Return-based 12-1 momentum uses cumulative returns from month $t-12$ through $t-2$. The most recent month is skipped to reduce short-term reversal contamination and to preserve conservative timing.

$$MOM_{i,t} = \prod_{j=2}^{12} (1 + r_{i,t-j}) - 1$$

Commodity and FX Momentum

Price-based 12-1 momentum is implemented through the equivalent lagged price ratio.

$$MOM_{i,t}^{px} = \frac{P_{i,t-2}}{P_{i,t-13}} - 1$$

Fixed-Income and FX Carry

Fixed-income carry is implemented as a pass-through of pre-computed yield/carry proxy series, with higher yield interpreted as higher carry. FX carry in the expanded version is implemented as the foreign 3-month rate minus the USD 3-month rate.

$$FXCarry_{i,t} = r^{\text{foreign}} 3M_{i,t} - r^{\text{USD}} 3M,$$

Sleeve	Signal	Rank direction (Long)	Economic interpretation
Equity Momentum	12-1 cumulative return	High is good	Country-level trend continuation
Commodity Momentum	$\frac{P(t-2)}{P(t-13)} - 1$	High is good	Commodity trend continuation
FI Carry	Yield/carry proxy	High is good	Compensation from yield, roll-down, and spread exposure
FX Momentum	$\frac{P(t-2)}{P(t-13)} - 1$	High is good	Currency trend continuation
FX Carry	Foreign 3M rate - USD 3M rate	High is good	Carry from holding higher-yielding currencies
FI Momentum	12-1 total return	High is good	Bond total-return trend continuation

For the above situations, the reverse holds true, for if shorting were needed.

2.2 Portfolio Construction

At each month-end for time t

1. Convert each sub-portfolio signal into cross-sectional rank-standardized raw weights of (as shown above):

$$w_{i,t}^{\text{raw}} = \frac{\text{rank}_{i,t} - \overline{\text{rank}_t}}{\sigma(\text{rank}_t)}$$

- Idea is that here, it will create long/short FMPs within each “sleeve”, such that the best-ranked assets receive positive weights, and vice versa.

2. Estimate a 36-month population covariance matrix using only information available at that date.

$$\hat{\Sigma}_t = 12 \times \text{Cov}(r_t^{-35}, \dots, r_t)$$

3. Scale each sleeve to 1% annualized ex-ante volatility.

$$w_t^{\text{scaled}} = w \times 0.01 / \sqrt{w_{\text{ra}} w_t^T \hat{\Sigma}_t w_{\text{ra}} w_t}$$

4. For the expanded strategy, the four momentum sleeves are equal-weighted within the Momentum family and rescaled to 1% annualized volatility. The two carry sleeves are equal-weighted within the Carry family and rescaled to 1% annualized volatility. The Momentum and Carry families are then equal-weighted and the final portfolio is rescaled again to a 1% annualized volatility target:

$$\begin{aligned}
w_t^{\text{MOM}} &= \frac{1}{4} w_t^{\text{EQ}} + \frac{1}{4} w_t^{\text{COM}} + \frac{1}{4} w_t^{\text{FI-MOM}} + \frac{1}{4} w_t^{\text{FX-MOM}} \xrightarrow{\text{scale}} 1\% \\
w_t^{\text{CARRY}} &= \frac{1}{2} w_t^{\text{FI-CARRY}} + \frac{1}{2} w_t^{\text{FX-CARRY}} \xrightarrow{\text{scale}} 1\% \\
w_t^{\text{final}} &= \frac{1}{2} w_t^{\text{MOM}} + \frac{1}{2} w_t^{\text{CARRY}} \xrightarrow{\text{scale}} 1\%
\end{aligned}$$

Note: Look-ahead bias was specifically mitigated and controlled through the pipeline, though signal timing by momentum skipping the most recent month, covariance matrix constructed for the last 36 months from time $-t$, and weights determined at time t are implemented at $t + 1$. A [notebook](#) was dedicated to auditing this.

3. Backtest Summary

The expanded backtest contains 240 monthly realized returns from December 2005 to November 2025, comfortably exceeding the minimum 10-year requirement. The baseline is better substantively on in-sample efficiency, but the expanded version is the stronger

GTA architecture, as it broadens the architecture set across fixed income and currencies, while keeping downside risks extremely similar.

The baseline performs better on in-sample efficiency with higher IR and lower turnover, but the expanded version is far more complete as it extends opportunity set across fixed income and currencies while keeping maximum drawdown almost unchanged.

The Carry family is the cleaner contributor (IR 0.33, Max DD -3.4%). FI Momentum (IR -0.075) and FX Carry (IR -0.012) individually detract over this sample, pulling the Momentum family IR to 0.12 and the aggregate IR to 0.36.

The RHS figure shows growth of \$1 for the final GTAA portfolio and benchmark comparison, together with the realised drawdown profile. The expanded strategy remains close to the baseline in cumulative performance and has a similar maximum drawdown portfolio.

The figure reports the growth of \$1 for the final portfolio and benchmark comparison, alongside the realised drawdown profile. Because the dataset begins in 2004 and the strategy uses three years of backward-looking covariance matrices, portfolio results begin around 2007. Overall, the expanded strategy remains close to the baseline in cumulative performance and has a similar maximum drawdown, suggesting that adding extra sleeves does not materially worsen tail risk.

The right-hand-side sleeves show the main incremental value and cost of the expansion. Commodity momentum is cyclical, performing well during parts of the post-2009 recovery and near the end of the sample, which is broadly consistent with commodity trend-following behaviour. FX momentum is largely flat over the full sample, with brief positive bursts but sharp drawdowns around 2016–2020 before a partial recovery. FX carry is the weakest sleeve, staying negative for much of the sample, consistent with poor realised performance of classical currency carry in low-rate regimes. The shaded stress periods help explain why the expanded strategy has higher turnover and lower risk-adjusted performance than the baseline, despite maintaining a similar maximum drawdown.

The *annual attribution chart* (shown in the visualisation gallery) confirms the diversification logic of the expanded strategy: negative contributions from one sleeve are frequently partially offset by positive contributions from another, though this does not guarantee higher realised performance when multiple sleeves simultaneously underperform. *Rolling 24-Month Correlation* shows that the correlation remains at about 0.13, being negative like in 2022 to 2024, the correlation becomes negative, which is what is desired.

Extended sleeve attribution charts, rolling IR, rolling volatility vs. target, and signal-level diagnostics are available in the [GitHub Visualization Gallery](#).

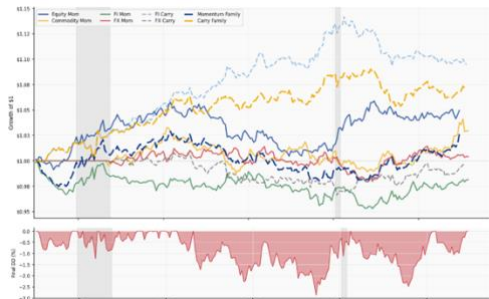
Expanded Strategy: Family-Level Performance

Sleeve or Family	Ann. return	Ann. vol	IR	Max DD	Avg DD	Ann. Turnover
Cross-Asset Momentum Family	0.13%	1.09%	0.123	-4.72%	-2.29%	22.4%
Cross-Asset Carry Family	0.35%	1.07%	0.326	-3.36%	-0.92%	7.3%
Expanded Final GTAA	0.42%	1.15%	0.360	-2.86%	-1.01%	19.4%

Expanded Strategy: Sleeve-Level Performance

Sleeve or Family	Ann. return	Ann. vol	IR	Max DD	Avg DD	Ann. Turnover
Equity Momentum FMP	0.40%	1.08%	0.374	-5.00%	-1.25%	9.1%
Commodity Momentum FMP	0.14%	0.98%	0.143	-4.39%	-2.34%	2.9%
Fixed-Income Momentum FMP	-0.07%	0.99%	-0.074	-4.96%	-2.71%	15.5%
FX Momentum FMP	0.02%	0.87%	0.027	-3.43%	-1.08%	13.7%
Fixed-Income Carry FMP	0.13%	0.58%	0.217	-4.21%	-1.32%	1.7%
FX Carry FMP	-0.01%	0.88%	-0.012	-4.13%	-1.96%	1.8%

Expanded Strategy — All 6 Sleeves & 2 Factor Families (Growth of \$1)

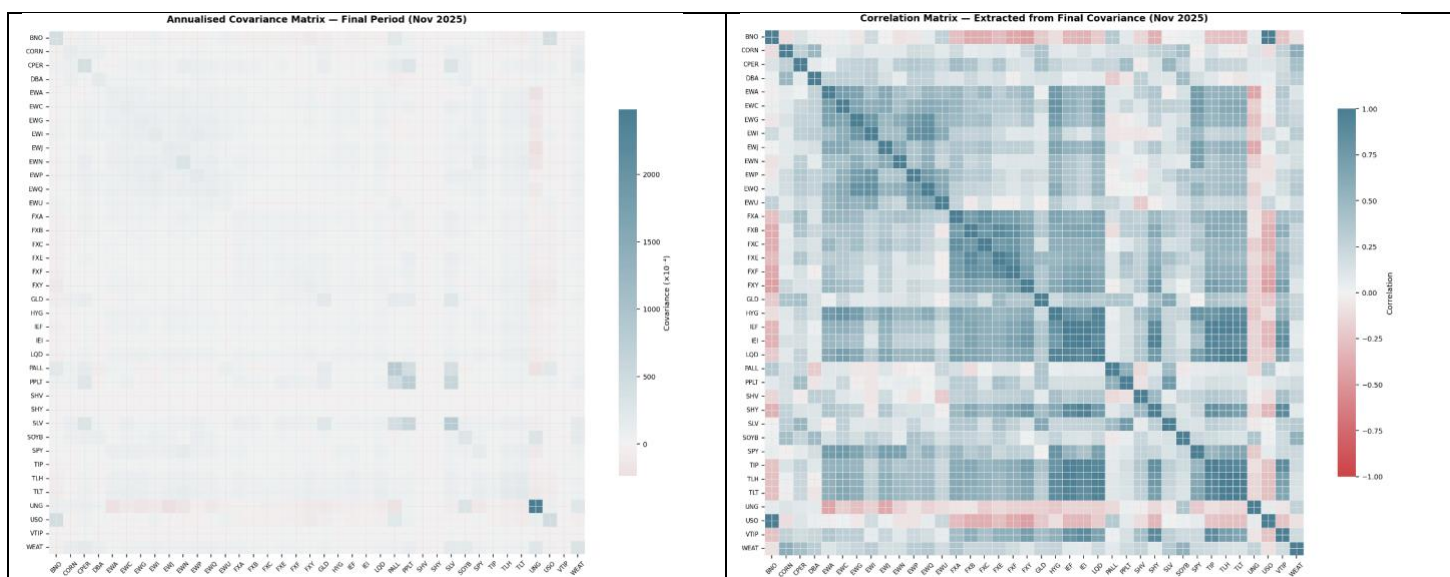


Growth of \$1 — Individual Sleeves (1% vol target each)



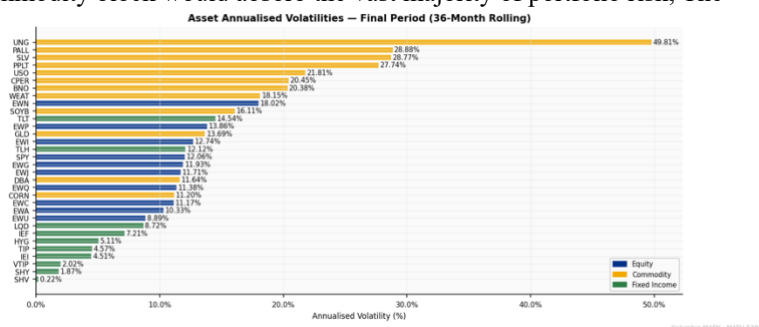
Looking at the final two years of the backtest across the weight panels, volatility bar chart, and correlation and covariance heatmaps, the portfolio's positioning tells a coherent macro story that maps closely onto what played out in markets during 2024 and into 2025.

On the equity side, the strategy spent most of the period persistently long Spain (+3.3%) and Italy (+3.2%) and short France (-3.7%) and Australia (-1.9%). European periphery markets held up well as ECB rate cuts and improving fiscal credibility supported them, while France was dragged lower by political uncertainty following the snap elections (Macron losing afterwards), and Australia underperformed as China demand remained soft. The US (SPY) sat in mild positive territory (+2.3%), consistent with the AI-driven outperformance of US equities across the window.



The fixed income component generated the most interesting positioning, with T-bills around 5% during the Fed's holding period, SHV ranked highly on carry (+3.4%) and held a persistent long alongside LQD (+6.6%), whose Baa yield proxy remained elevated in absolute terms. The short side was dominated by HYG (-5.9%) and VTIP (-5.5%): high-yield credit spreads were historically tight, depressing HYG's carry ranking despite high nominal yields, while short-maturity TIPS ranked poorly as compressed real yields left them at the bottom of the cross-section.

This volatility bar chart really showcases the importance of decomposition, wherein UNG, PALL and SLV sit far above every other asset, confirming that without sleeve-level vol targeting, the commodity block would absorb the vast majority of portfolio risk. The correlation heatmap, shows that intra-class-wise, equity ETFs are moderately correlated (0.5-0.9), treasury ETFs are clustered by duration between (0.7 – 0.95), and commodity ETFs are largely uncorrelated with each other, and other asset classes. These near-zero cross-class correlations are the structural reason the combined portfolio achieves a maximum drawdown of -2.83%, well inside any individual sleeve's worst drawdown of -4.2% to -5.0%.



The final covariance, volatility, and correlation outputs are reported for the final rebalance period used in the backtest, November 2025. Additional final-period outputs are available in the [final submission notebook](#) and the [GitHub repository](#) (like Dec-2025 figures, as the guide was not that clear).

5. Conclusion

In conclusion, this project constructs and backtests a long/short factor-based GTAA portfolio using a disciplined two-family framework built around cross-asset momentum and carry. The final expanded specification broadens the opportunity set across equities, commodities, fixed income, and FX, while preserving the same monthly rebalancing, 36-month covariance estimation, 1% sleeve volatility targeting, and no-look-ahead implementation logic required by the assignment. Although the baseline version delivers slightly stronger in-sample return efficiency, the expanded portfolio is the more complete GTAA design because it applies the same factor construction consistently across a wider cross-section of global asset classes. There are notable limitations, shown [here](#).

Overall, the results suggest that the expanded architecture provides a reasonable and methodologically sound framework for diversified tactical allocation, with performance driven more by the robustness of the construction process than by any single sleeve.

Full methodology, data tables, and construction-level code are available in the [technical report](#) and [repository](#).